Christopher Lee

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Education

Columbia University New York, NY

B.A. Mathematics & Computer Science — GPA: 3.94 Aug 2025 – Present

Coursework: Probability Theory, Modern Analysis, Modern Algebra, Machine Learning, Advanced Programming in C, Discrete Math, Fundamentals of Computer Systems, Multi-variable Calculus, Linear Algebra, Data Structures in Java

Experience

Columbia University Math Department Research Program New York, NY

CSUREMM Researcher *May 2026 – Aug 2026*

- Developed a framework for adaptive verification in AI reasoning graphs, using a σ-threshold risk policy to optimize uncertainty-aware decisions under constrained token budgets
- Built an LLM pipeline in Python to generate and verify 5,000+ number-theory claims across 8 LLMs, measuring per-model calibration and error propagation
- Co-authored paper, research project sponsored by Jane Street, presenting at the Columbia Research Symposium and Jane Street Summer Math Day

QandQ AI New York, NY

Quantitative Modeling Intern *Jan 2026 – Aug 2026*

- Traced and documented a production Jenkins-to-Redis pipeline generating AI valuation data for 78+ tickers
- Designed a PostgreSQL schema, then implemented store_to_pg() to persist LangGraph pipeline output alongside Redis

MeiraGTx New York, NY

Machine Learning Intern *Jul 2025 – Aug 2025*

- Engineered Python data pipeline to process 10M+ nucleotide sequences; applied PCA and ML models to inform data-driven capsid selection decisions
- Engineered AAV capsids via transduction/transfection experiments; analyzed potency using qPCR and high-throughput screening at biotech gene therapy startup

Columbia University Computer Science Department New York, NY

COMS W3203 Discrete Mathematics Teaching Assistant *Sep 2026 – Present*

- Lead weekly Friday recitations and office hours, covering proofs, graph theory, combinatorics, probability, and number theory
- Grade weekly handwritten problem sets on Gradescope and provide written feedback on proof technique and mathematical communication

Application Development Initiative New York, NY

Media Committee *Feb 2026 – Present*

- Marketing and newsletter for Columbia's largest computer science club (2000+ members), organize and promote hackathons, workshops, speaker events

Projects

Structure-Aware Verification Budget Allocation in LLM Reasoning [paper] *May 2026 – Aug 2026*

- Modeled LLM reasoning as dependency DAGs and designed structure-aware verification strategies under a fixed token budget, benchmarking across 8 open- and closed-source LLMs
- Built a Python pipeline to query LLM APIs (Groq, OpenRouter) at scale, generating and auto-verifying thousands of test claims with retry logic and robust parsing for malformed model outputs

Lionplan | *ADI CodeCollab Hackathon* *Apr 2026 – May 2026*

- Built a full-stack academic planning platform (Next.js, React, TypeScript) with a drag-and-drop 8-semester course planner, syncing to a Supabase/PostgreSQL backend for real-time Core and major requirement tracking
- Engineered Python scraping pipeline (Playwright, BeautifulSoup) to read live section data and professor reviews from Columbia's SIS Directory and CULPA, upsert into Supabase

Skills and Awards

Programming Languages/Tools: Java, Python, C, LaTeX, React, SQL, Excel, Git/GitHub, Docker

Awards: 3x AIME Qualifier, Dean's List, National Merit Scholar